Practice Session 4:

Working with PAAS (Azure App Service)

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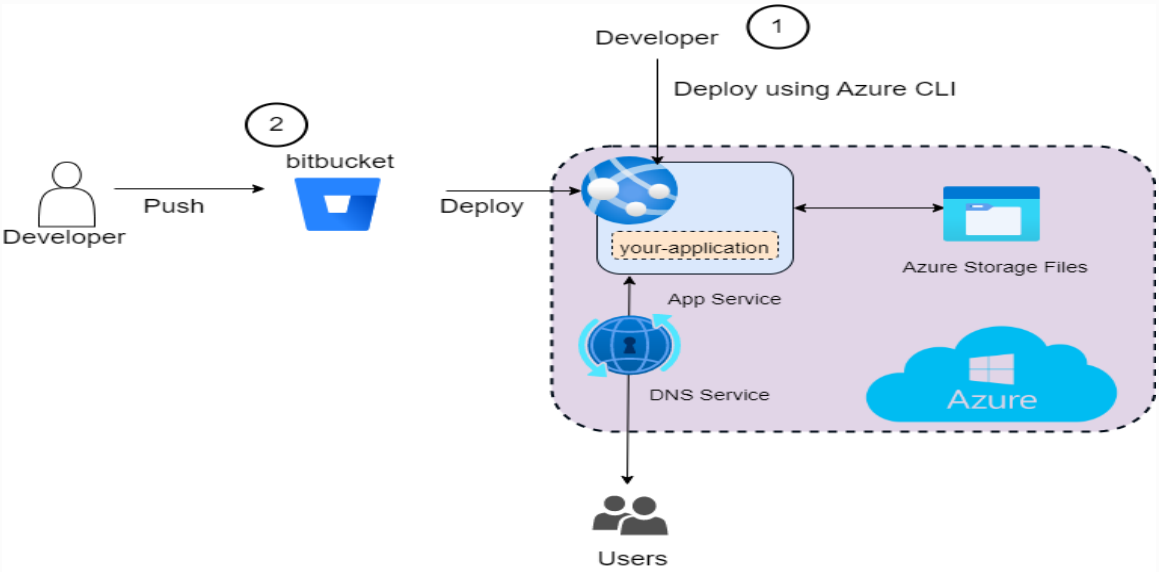
Practice 4: Platform as a Service

During this practice session, we'll look into Microsoft Azure's Platform as a Service (PaaS). We'll focus on deploying web applications as PaaS offerings and examining various features of this cloud platform.

Applications can be deployed onto PaaS in Multiple ways:

Using CLI (Command Line Interface) - Manually deploying the code

1. Git-based (BitBucket, GitHub) - Linking code from your app repository
2. Using docker image registries - Linking a Docker image from a container registry



References

1. Azure Cloud documentation: https://docs.microsoft.com/en-us/azure/?product=featured
2. Azure App Service: https://learn.microsoft.com/en-us/azure/app-service/
3. Azure Storage: https://learn.microsoft.com/en-us/azure/storage/common/storage-introduction

Exercise 4.1. Getting familiar with the Azure platform and deploying to App Service using docker images

Task 4.1.1: Build and push the docker image of the flask message-board application.

- We need a docker-machine to perform this task. (use the WSL/ubuntu installed in las exp on your desktop)
- Fork the project to your Bitbucket or GitHub account. (We will update the source code in further exercises)
    - Repos to fork (You can use either of one based on your choice):

        - GitHub repo:
        - PS!! Make your repository private.
- Clone your project to a local machine or VM
- Build and push the image to the docker hub
        - It's nice to test the image locally in the VM or your system before pushing it to the Docker hub. This is to avoid run time errors while deploying container image into Azure App Service.
    - NB! if you build in MacOS, make sure to build the Docker image for Linux/amd64 by adding `--platform linux/amd64` to the build command

Task 4.1.2: Deploying docker image to Azure App Service.

- Log into Azure cloud: https://azure.microsoft.com/en-us/
    - Use your university email and password
    - If asked which subscription type to use, select the one provided by university
- Open the Azure Cloud portal: https://portal.azure.com/#home
- Familiarize yourself with the available services: https://portal.azure.com/#allservices

Azure App Service is a fully managed platform for building web applications. In this task, we will deploy the message-board application into the App service.

- Search for **App Services** in the azure portal.
- Click on **Create** --> **Web App**
    - Subscription- Choose **Azure for students**
    - Choose **Resource Group --> Create New--> Name** :
    - **Instance Details**
        - Choose a name freely (don't use any special characters)
        - Select Publish --> Docker Container
        - Region --> North Europe
        - Pricing plan : ----------
- Click on Next: Database (Nonthing do here)
- Click on Next: Docker
    - Image Source --> Docker Hub
    - Docker hub options
        - Image and tag --> your-dockerhub-id/appname
- Click on **Review and Create** and than **Create**.
- Now, it will take a few minutes to deploy the application and click on **Go to Resource**
    - If errors occur during deployment, such as Error 429, it's advisable to cancel the deployment and start fresh by selecting the **Region --> West Europe** option.
- Go to Overview of your app service and copy the URL (**Essentials-->URL**) and open your application in the browser.
    - It will take a few minutes for the application to run (Ex: 5 to 6 min for me).
    - You can see the deployment logs of the application service **Deployment**--> **Deployment Center**-->**View Logs**

- You can see the container logs **Monitoring**--> **Log Stream**

- Once your application runs, test by inserting a few messages.
- Keep the application running and don't access it until you reach Ex 3.5 (PS!! You can continue with the next exercise, and this is to test the data persistence)